Srihari Vemuru

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SKILLS

Perception & Vision: OpenCV, Open3D, YOLOv11, CenterNet, PyTorch, Tensorflow, Point Cloud Processing, NeRF
SLAM & State Estimation: ORB-SLAM3, EKF/UKF Sensor Fusion, GPS+IMU Integration, LiDAR/Camera Calibration, Occupancy Mapping
Robotics Middleware & Simulation: ROS, ROS2, Isaac Sim, MuJoCo, Gazebo, CARLA, SUMO, Unreal Engine, Docker
Languages & Systems: C/C++, Python, MATLAB, Linux/Unix, Bash, File Systems

EXPERIENCE

Iowa State University **August 2024 - Present**
Research Assistant *Ames, IA, USA*
- Implemented ORB-SLAM3-based visual localization on an autonomous ground delivery robot in ROS2, achieving stable map tracking and drift-bounded pose estimation in dynamic outdoor environments.
- Developed a real-time YOLOv11 pedestrian detection pipeline for deployment on autonomous delivery robots, enabling reliable obstacle identification at inference speeds suitable for onboard navigation.
- Built a co-simulation environment in CARLA and Unreal Engine to model dynamic urban traffic conditions, providing a reproducible testbed for evaluating robot perception and navigation stacks.

Strider Robotics **September 2022 - July 2024**
Robotics AI Engineer *Bengaluru, India*
- Built a multi-robot state estimation framework using GPS, IMU, and EKF-based sensor fusion to maintain unified world-frame localization across heterogeneous robotic fleets of 2 UAVs and a Unitree GO1 quadruped.
- Developed a navigation framework using A*, ROS2, and NVIDIA Isaac Sim, enabling collision-free quadruped navigation in dynamic agricultural environments.
- Designed and fabricated custom 3D-printed mounts for sensors, cameras, and agricultural sprayers, enabling system integration and deployment of autonomous UAV and quadruped robotic systems.

LightMetrics **May 2020 - August 2020**
AI Intern *Bengaluru, KA, India*
- Integrated CenterNet with YOLO for road sign and vehicle detection, improving multi-class detection coverage for on-road scenarios.
- Applied CNN pruning techniques to optimize on-road detection models for deployment, reducing computational overhead while maintaining competitive mAP on test datasets.

EDUCATION

Iowa State University **August 2024 - December 2026**
Master of Science (MS), Mechanical Engineering *GPA: 3.72*

International Institute of Information Technology Bangalore, India **August 2020 - July 2022**
Master's, Computer Science *GPA: 3.42*

International Institute of Information Technology Bangalore, India **August 2017 - July 2020**
Bachelor's, Computer Science *GPA: 3.42*

PROJECTS

NeRFly **January 2025 - Present**
Research Assistant at Iowa State University *Ames, IA, USA*
- Built an autonomous UAV reconstruction pipeline in Gazebo Harmonic with PX4 SITL and MAVROS2, capturing 500-image datasets via autonomous orbit missions in OFFBOARD mode.
- Benchmarked COLMAP SfM, nerfacto, and 3D Gaussian Splatting reconstructions against Gazebo ground truth mesh using Chamfer distance and F-score, achieving 0.60px mean reprojection error across 100 registered images.
- *Link to project*

TerraPPO **August 2022 - June 2024**
Robotics Engineer at Strider Robotics *Bengaluru, KA, India*
- A MuJoCo-based quadruped simulation using Unitree Go1 URDF with terrain randomization (slopes, steps, deformable surfaces), training PPO agents for robust locomotion and improved sim-to-real transfer.
- *Link to project*

PUBLICATIONS

Srihari Vemuru, Kumar Ankit, Sarvesh Prasanth Venkatesan, Debashish Ghose, Shishir Kolathaya. Yield Prediction and Counting Using World Coordinates. In *ASME IMECE-India 2026 Proceedings*. ASME.